

Workshop: Matrix-Vector Equations

**Geometric Algorithms
Lecture 5**

Recall: The Fundamental Connection

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix}$$

augmented matrix

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

system of linear equations

$$x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

vector equation

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augmented matrix

Why not view
these as a
vector too?

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vector equation

Matrix-Vector Multiplication

Matrix-Vector "Interface"

multiplication

what does $A\mathbf{v}$ mean when A is a matrix and $\mathbf{v}$ is a vector?

Matrix-Vector Multiplication (Pictorially)

AS

Matrix-Vector Multiplication (Pictorially)

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{bmatrix}$$

Matrix-Vector Multiplication (Pictorially)

The diagram illustrates the process of matrix-vector multiplication. On the left, a matrix is shown with its columns highlighted in red. The columns are labeled a_{11} , a_{12} , ..., a_{1n} for the first row, a_{21} , a_{22} , ..., a_{2n} for the second row, $\vdots$ for the middle rows, and a_{m1} , a_{m2} , ..., a_{mn} for the last row. To the right of the matrix is an ellipsis $\dots$. Further right is a vector with its elements highlighted in blue, labeled s_1 , s_2 , $\vdots$, and s_n . The entire expression is enclosed in large square brackets.

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{bmatrix}$$

Matrix-Vector Multiplication (Pictorially)

$$s_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + s_2 \begin{bmatrix} a_{21} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \dots + s_n \begin{bmatrix} a_{m1} \\ a_{m2} \\ \vdots \\ a_{mn} \end{bmatrix}$$

Matrix-Vector Multiplication (Pictorially)

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a linear combination of the columns where
s defines the weights

Why keeping track of matrix size is important

this only works if the number of *columns* of the matrix matches the number of *rows* of the vector

$$\begin{bmatrix} * & \dots & * \\ * & \dots & * \\ \vdots & \ddots & \vdots \\ * & \dots & * \\ * & \dots & * \end{bmatrix} \begin{bmatrix} * \\ \vdots \\ * \end{bmatrix} = \begin{bmatrix} * \\ * \\ \vdots \\ * \\ * \end{bmatrix}$$

$(m \times n) \qquad (n \times 1) \qquad (m \times 1)$

Why keeping track of matrix size is important

this only works if the number of *columns* of the matrix matches the number of *rows* of the vector

$$\begin{array}{ccc} \begin{array}{c} \textcolor{blue}{m} \end{array} \begin{array}{c} \textcolor{red}{n} \\ \left[\begin{array}{ccc} * & \dots & * \\ * & \dots & * \\ \vdots & \ddots & \vdots \\ * & \dots & * \\ * & \dots & * \end{array} \right] \end{array} & \begin{array}{c} \textcolor{red}{n} \begin{array}{c} \textcolor{purple}{1} \\ \left[\begin{array}{c} * \\ \vdots \\ * \end{array} \right] \end{array} \end{array} = \begin{array}{c} \textcolor{blue}{m} \begin{array}{c} \textcolor{purple}{1} \\ \left[\begin{array}{c} * \\ * \\ \vdots \\ * \\ * \end{array} \right] \end{array} \end{array} \\ \begin{array}{c} (\textcolor{blue}{m} \times \textcolor{red}{n}) \end{array} & \begin{array}{c} (\textcolor{red}{n} \times \textcolor{purple}{1}) \end{array} & \begin{array}{c} (\textcolor{blue}{m} \times \textcolor{purple}{1}) \end{array} \end{array}$$

Non-Example

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 4 \end{bmatrix} + 3 ???$$

Non-Example

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Non-Example

The diagram illustrates a matrix multiplication operation. On the left, a 2×2 matrix (represented by two red vertical rectangles) is multiplied by a 3×1 vector (represented by a blue vertical rectangle with three cells). The matrix contains the values 1 and 2 in the top row, and 3 and 4 in the bottom row. The vector contains the values 1, 2, and 3. This is followed by an equals sign and a sum of three terms. The first term is a scalar 1 (in a blue box) multiplied by a 2×1 vector (red rectangle) containing 1 and 3. The second term is a scalar 2 (in a blue box) multiplied by a 2×1 vector (red rectangle) containing 2 and 4. The third term is a scalar 3 (in a blue box) followed by three question marks, indicating an unknown result.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 4 \end{bmatrix} + 3 ???$$

$(2 \times 2) \quad (3 \times 1)$

Non-Example

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THESE DON'T MATCH

(2×2) (3×1)

Example

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THESE MATCH
 (2×2) (2×1)

Matrix-Vector Multiplication

Df. Given a $(m \times n)$ matrix A with columns $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$, and a vector $\mathbf{v}$ in $\mathbb{R}^n$, we define

$$A\mathbf{v} = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_n] \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = v_1\mathbf{a}_1 + v_2\mathbf{a}_2 + \dots v_n\mathbf{a}_n$$

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$A\mathbf{v}$ is a linear combination of the columns of A with weights given by $\mathbf{v}$

Algebraic Properties

The algebraic properties of matrix–vector multiplication are **very important**

$$1. \quad A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v}$$

$$2. \quad A(c\mathbf{v}) = c(A\mathbf{v})$$

A Common Error

$$A\mathbf{v} \neq \mathbf{v}A$$

Important. We multiply matrices and vectors
with the matrix *on the left*

A Tip

$$\begin{bmatrix} 2 & -3 & 4 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 5 \\ 5 \\ 4 \end{bmatrix} = \begin{bmatrix} ? \\ ? \end{bmatrix}$$

A Tip

$$\begin{bmatrix} 2 & -3 & 4 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 5 \\ 5 \\ 4 \end{bmatrix} = \begin{bmatrix} ? \\ ? \end{bmatrix}$$

$$5(2) + 5(-3) + 4(4) = 11$$

A Tip

$$\begin{bmatrix} 2 & -3 & 4 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 5 \\ 5 \\ 4 \end{bmatrix} = \begin{bmatrix} 11 \\ ? \end{bmatrix}$$

$$5(-1) + 5(1) + 4(0) = 0$$

A Tip

$$\begin{bmatrix} 2 & -3 & 4 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 5 \\ 5 \\ 4 \end{bmatrix} = \begin{bmatrix} 11 \\ 0 \end{bmatrix}$$

Calculating $A\mathbf{v}$

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{bmatrix} = \begin{bmatrix} ? \\ ? \\ \vdots \\ ? \end{bmatrix}$$

$$v_1 = a_{11}s_1 + a_{12}s_2 + \dots + a_{1n}s_n = \sum_{i=1}^n a_{1i}s_i$$

Calculating $A\mathbf{v}$

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{bmatrix} = \begin{bmatrix} v_1 \\ ? \\ \vdots \\ ? \end{bmatrix}$$

$$v_2 = a_{21}s_1 + a_{22}s_2 + \dots + a_{2n}s_n = \sum_{i=1}^n a_{2i}s_i$$

Calculating $A\mathbf{v}$

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ ? \end{bmatrix}$$

$$v_m = a_{m1}s_1 + a_{m2}s_2 + \cdots + a_{mn}s_n = \sum_{i=1}^n a_{mi}s_i$$

Calculating $A\mathbf{v}$

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_m \end{bmatrix}$$

Row-Column Rule

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^n a_{1i}s_i \\ \sum_{i=1}^n a_{2i}s_i \\ \vdots \\ \sum_{i=1}^n a_{mi}s_i \end{bmatrix}$$

Inner product: $[a_1 \ a_2 \ \cdots \ a_n] \begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{bmatrix} = \sum_{i=1}^n a_i s_i$

Inner Product

Df. The **inner product** of vectors **u** and **v** in $\mathbb{R}^n$ is defined the

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^n u_i v_i$$

Row-Column Rule

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^n a_{1i}s_i \\ \sum_{i=1}^n a_{2i}s_i \\ \vdots \\ \sum_{i=1}^n a_{mi}s_i \end{bmatrix}$$

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Row-Column Rule

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The i th entry of the As is the inner product of the i th row of A and s

The Matrix Equation

Recall: Vector Equations

$$x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \dots + x_n \mathbf{a}_n = \mathbf{b}$$

Recall: Vector Equations

$$x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \dots + x_n \mathbf{a}_n = \mathbf{b}$$

Q. Can $\mathbf{b}$ be written as a linear combination of $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$?

Recall: Vector Equations

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A. think of the weights as *unknowns*

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Q. Can $\mathbf{b}$ be written as a linear combination of $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$?

A. think of the weights as *unknowns*

we can use the same idea for matrix–vector
multiplication

The Matrix Equation

$$A\mathbf{x} = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_n] \mathbf{x} = \mathbf{b}$$

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$$A\mathbf{x} = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_n] \mathbf{x} = \mathbf{b}$$

Can $\mathbf{b}$ be written as a linear combination of **the columns of A** ?

The Matrix Equation

$$A\mathbf{x} = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_n] \mathbf{x} = \mathbf{b}$$

Can $\mathbf{b}$ be written as a linear combination of **the columns of A** ?

The Idea. write the "vector part" of our matrix-vector multiplication as an *unknown*

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

How To: The Matrix Equation

Q. Does $A\mathbf{x} = \mathbf{b}$ have a solution?

Q. Is $A\mathbf{x} = \mathbf{b}$ consistent?

Q. Determine a general form solutions of $A\mathbf{x} = \mathbf{b}$

How To: The Matrix Equation

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A. We can write this as:

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A. We can write this as:

(matrix equation)

$$[\mathbf{a}_1 \quad \mathbf{a}_2 \dots \mathbf{a}_n] \mathbf{x} = \mathbf{b}$$

How To: The Matrix Equation

Q. Determine the general form solution of $A\mathbf{x} = \mathbf{b}$

A. We can write this as:

(matrix equation)

$$[\mathbf{a}_1 \quad \mathbf{a}_2 \dots \quad \mathbf{a}_n] \mathbf{x} = \mathbf{b}$$

(vector equation)

$$x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \dots x_n \mathbf{a}_n = \mathbf{b}$$

How To: The Matrix Equation

Q. Determine the general form solution of $A\mathbf{x} = \mathbf{b}$

A. We can write this as:

(matrix equation)

$$[\mathbf{a}_1 \quad \mathbf{a}_2 \dots \mathbf{a}_n] \mathbf{x} = \mathbf{b}$$

(vector equation)

$$x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \dots x_n \mathbf{a}_n = \mathbf{b}$$

They both have the same solution set

How To: The Matrix Equation

Q. Determine the general form solution of $A\mathbf{x} = \mathbf{b}$

A. Use Gaussian elimination (or other means) to convert $[\mathbf{a}_1 \ \mathbf{a}_2 \ \dots \ \mathbf{a}_n \ \mathbf{b}]$ to reduced echelon form

Full Span

Recall: Span

Recall: Span

Df. the *span* of a set of vectors is the set of all possible linear combinations of them

$$\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} = \{\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots \alpha_n \mathbf{v}_n : \alpha_1, \alpha_2, \dots, \alpha_n \text{ are in } \mathbb{R}\}$$

Recall: Span

Df. the *span* of a set of vectors is the set of all possible linear combinations of them

$$\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} = \{\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots \alpha_n \mathbf{v}_n : \alpha_1, \alpha_2, \dots, \alpha_n \text{ are in } \mathbb{R}\}$$

$\mathbf{u} \in \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ exactly when $\mathbf{u}$ can be expressed as a linear combination of those vectors

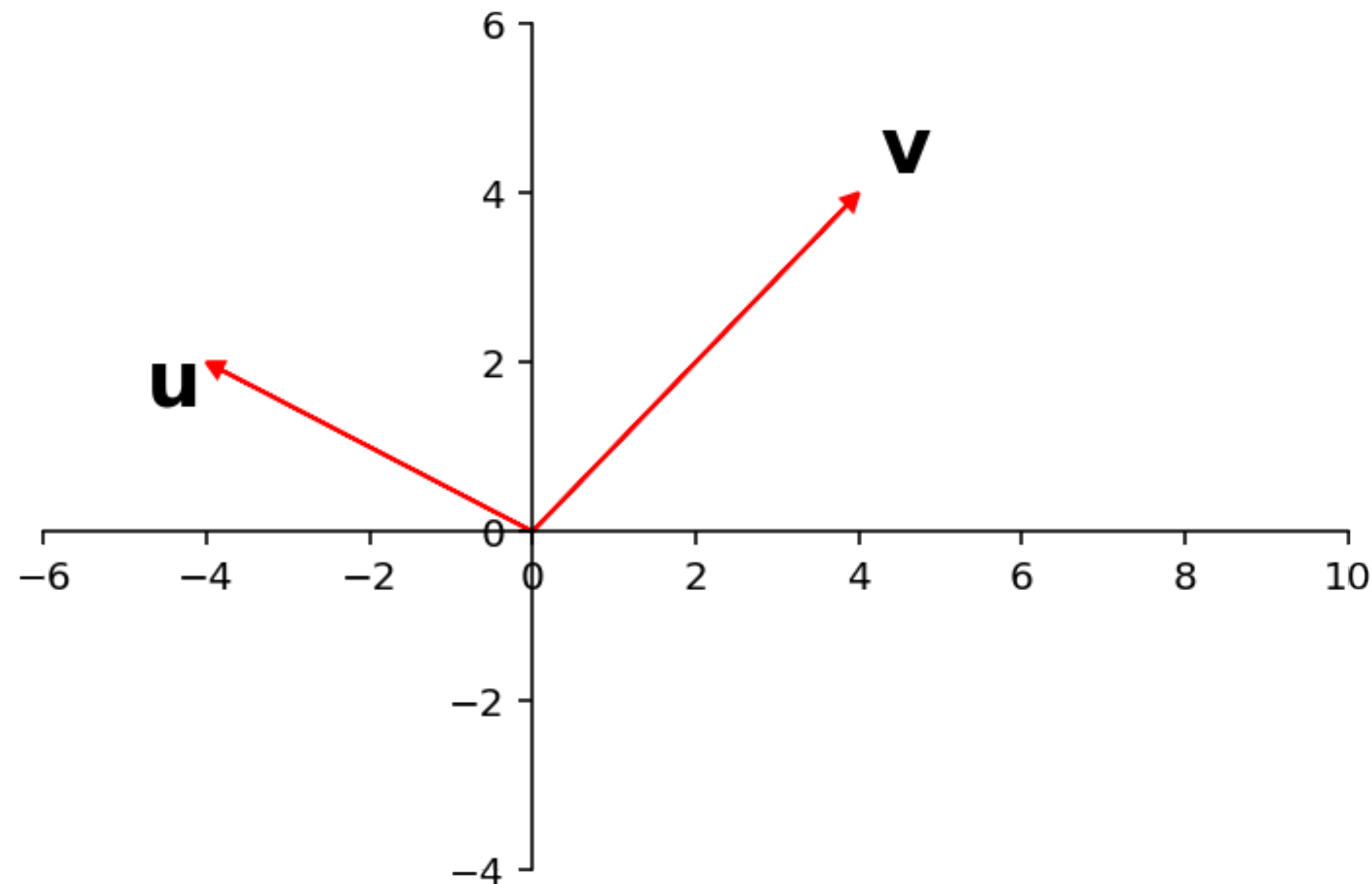
Spanning all of $\mathbb{R}^2$

Spanning all of $\mathbb{R}^2$

If two (or more) vectors in $\mathbb{R}^2$ span a plane, they must span all of $\mathbb{R}^2$. They "fill up" $\mathbb{R}^2$

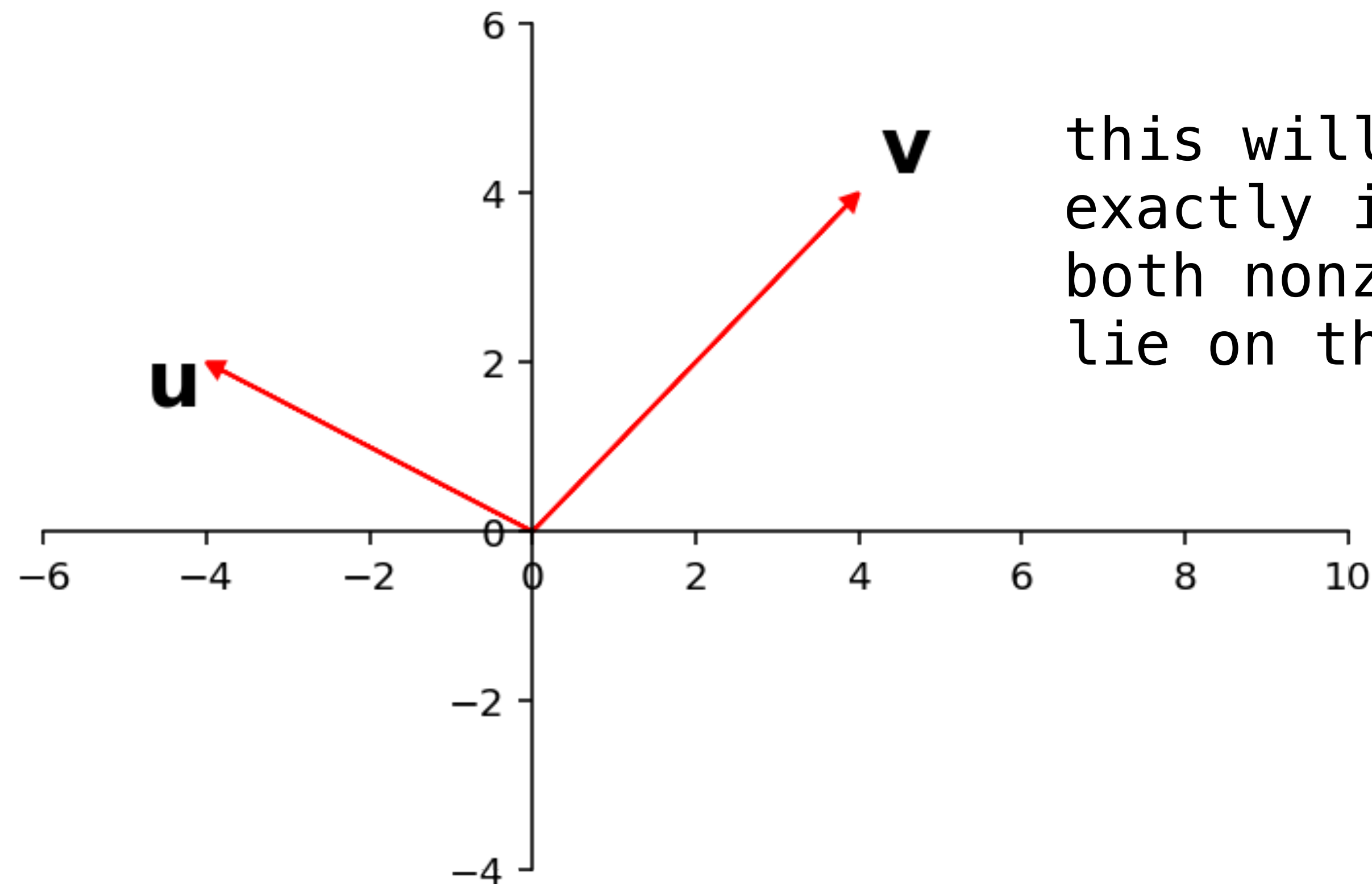
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Spanning all of $\mathbb{R}^2$

If two (or more) vectors in $\mathbb{R}^2$ span a plane, they must span all of $\mathbb{R}^2$. They "fill up" $\mathbb{R}^2$



this will happen
exactly if they are
both nonzero and don't
lie on the same line

What about $\mathbb{R}^n$?

When do a set of vectors span all of $\mathbb{R}^n$?

When do a set of vectors "fill up" $\mathbb{R}^n$?

A Thought Experiment

Suppose I give you the augmented matrix of a linear system but I cover up the last column

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & \blacksquare \\ 2 & 1 & 0 & \blacksquare \end{array} \right]$$

A Thought Experiment

Then we reduce it to echelon form

$$\begin{bmatrix} 1 & 2 & 3 & \blacksquare \\ 2 & 1 & 0 & \blacksquare \end{bmatrix}$$

A Thought Experiment

Then we reduce it to echelon form

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & \blacksquare \\ 2 & 1 & 0 & \blacksquare \end{array} \right]$$

$$R_2 \leftarrow R_2 - 2R_1$$

A Thought Experiment

then we reduce it to echelon form

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \end{bmatrix} \begin{array}{c} \blacksquare \\ \blacksquare \end{array}$$

A Thought Experiment

then we reduce it to echelon form

$$\begin{bmatrix} 1 & 2 & 3 & \blacksquare \\ 0 & -3 & -6 & \blacksquare \end{bmatrix}$$

Does it have a solution?

A Thought Experiment

then we reduce it to echelon form

$$\begin{bmatrix} 1 & 2 & 3 & \blacksquare \\ 0 & -3 & -6 & \blacksquare \end{bmatrix}$$

Yes. It doesn't have an inconsistent row

A Thought Experiment

what about this system?

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & 2 & 4 \end{bmatrix} \begin{bmatrix} \\ \end{bmatrix}$$

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$$R_2 \leftarrow R_2 - 2R_1$$

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it depends...

Pivots and Spanning $\mathbb{R}^m$

$$\begin{bmatrix} 1 & 2 & 3 & \blacksquare \\ 2 & 1 & 0 & \blacksquare \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 & \blacksquare \\ 0 & -3 & -6 & \blacksquare \end{bmatrix}$$

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If it doesn't matter what the last column is,
then **every choice must be possible**

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**Every vector in $\mathbb{R}^2$ can be written as a linear
combination of $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$, and $\begin{bmatrix} 3 \\ 0 \end{bmatrix}$**

Spanning $\mathbb{R}^m$

Theorem. For any $m \times n$ matrix, the following are logically equivalent

1. For every $\mathbf{b}$ in $\mathbb{R}^m$, $A\mathbf{x} = \mathbf{b}$ has a solution
2. The columns of A span $\mathbb{R}^m$
3. A has a pivot position in every row

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How To: Spanning $\mathbb{R}^m$

Q. Does the set of vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ from $\mathbb{R}^m$ span all of $\mathbb{R}^m$?

A. Reduce $[\mathbf{a}_1 \ \mathbf{a}_2 \ \dots \ \mathbf{a}_n]$ to echelon form and check if every row has a pivot

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!! We only need the echelon form !!

Not spanning $\mathbb{R}^m$

$$\begin{bmatrix} 1 & 1 & 2 & \blacksquare \\ 2 & 2 & 4 & \blacksquare \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 & \blacksquare \\ 0 & 0 & 0 & \blacksquare \end{bmatrix}$$

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We can't make the last column $[0 \ 0 \ 0 \ \blacksquare]$ for nonzero $\blacksquare$

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But we can make the last column parameters to find equations that must hold

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$$\begin{bmatrix} 1 & 1 & 2 & b_1 \\ 2 & 2 & 4 & b_2 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 & b_1 \\ 0 & 0 & 0 & b_2 - 2b_1 \end{bmatrix}$$

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**This gives use a linear equation which describes
the span of $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 4 \end{bmatrix}$**

Four Representations

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix}$$

augmented matrix

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

matrix equation

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

system of linear equations

$$x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

vector equation

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matrix equation

they all have the same solution sets

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vector equation

Workshop

What you need to be able to do:

- » Compute matrix-vector products
- » Solve matrix-vector equations
- » Determine if a set of vectors have full span
- » (other related questions)

Summary

Matrix and vectors can be multiplied together to get new vectors

The matrix equation is another representation of systems of linear equations

Looking forward: Matrices *transform* vectors